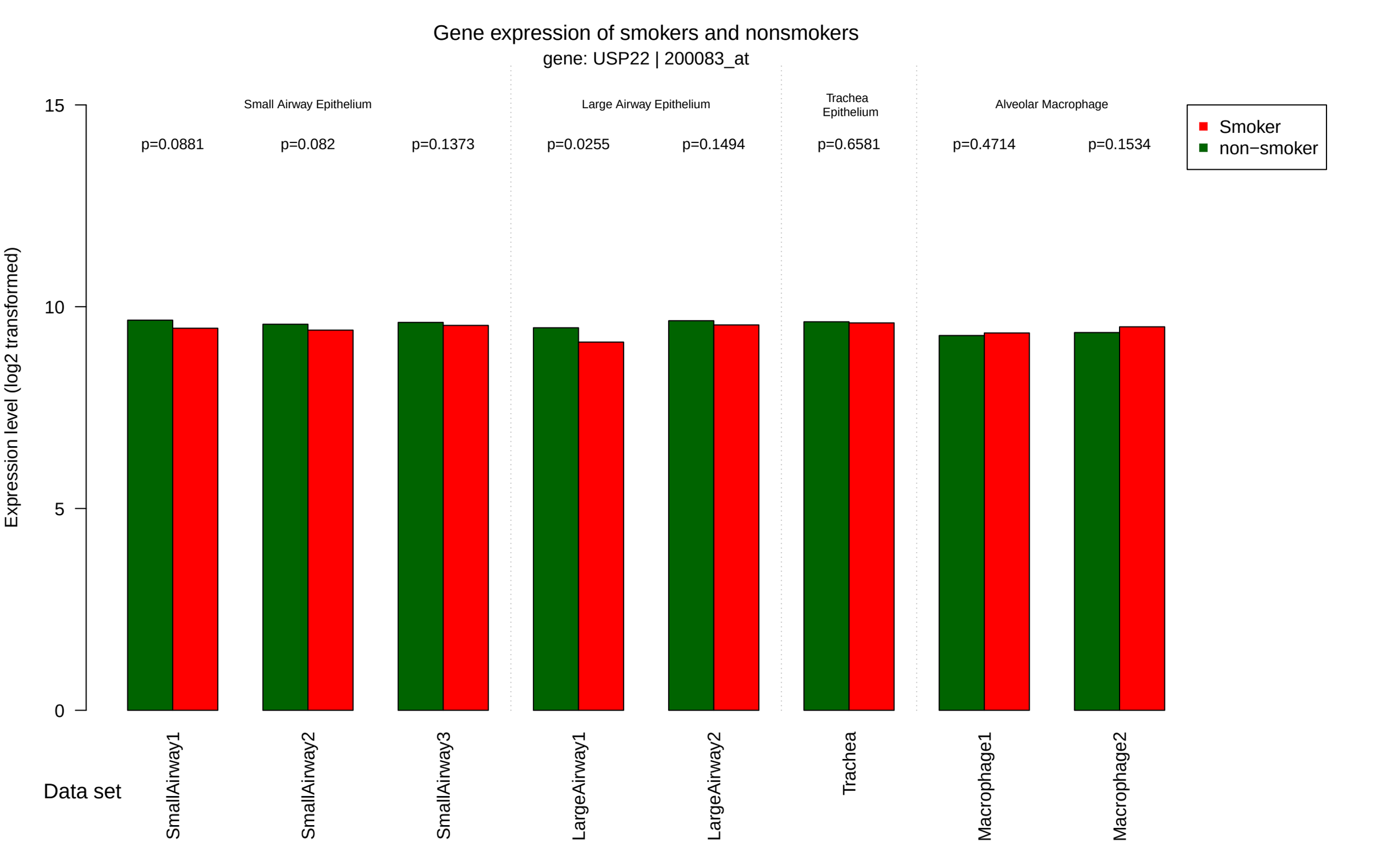
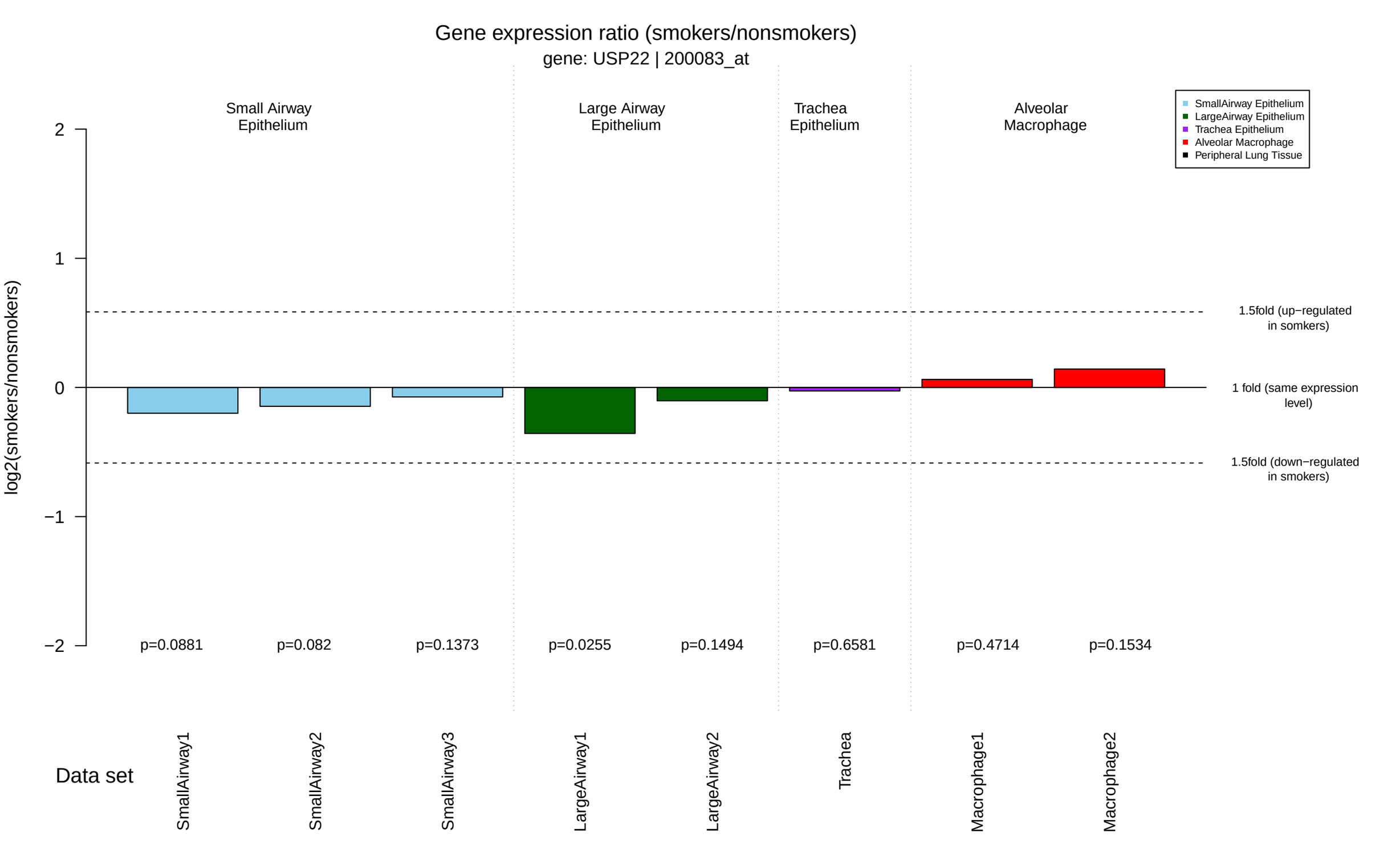
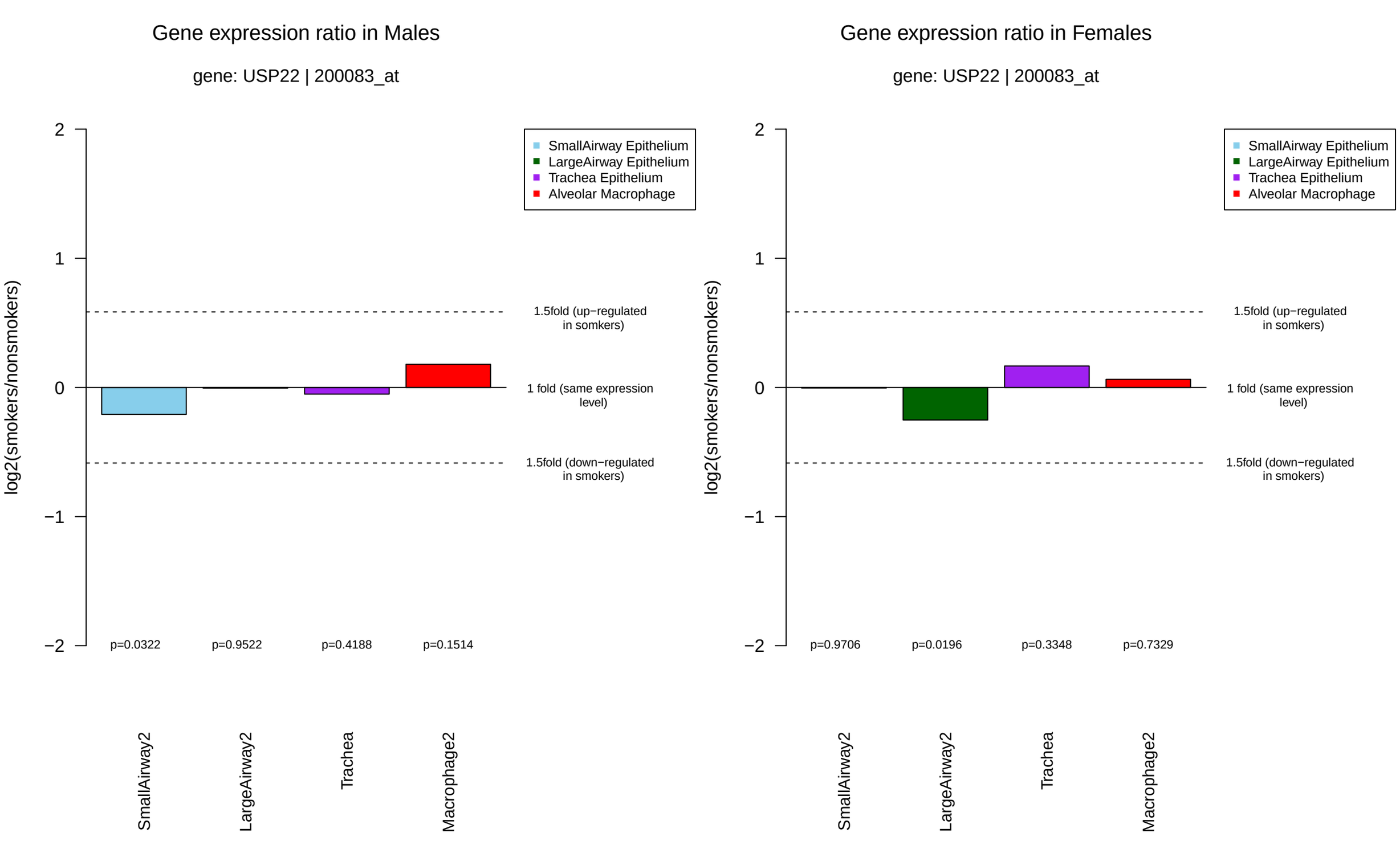
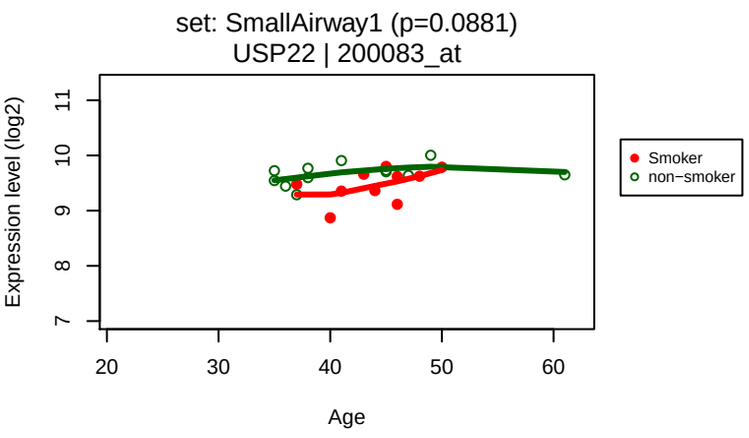
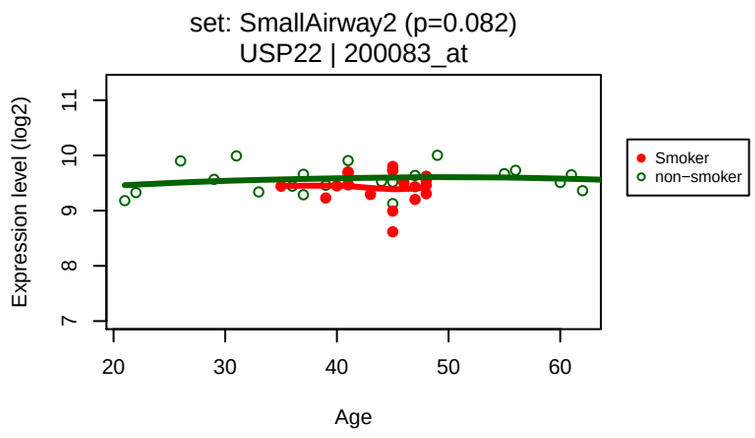
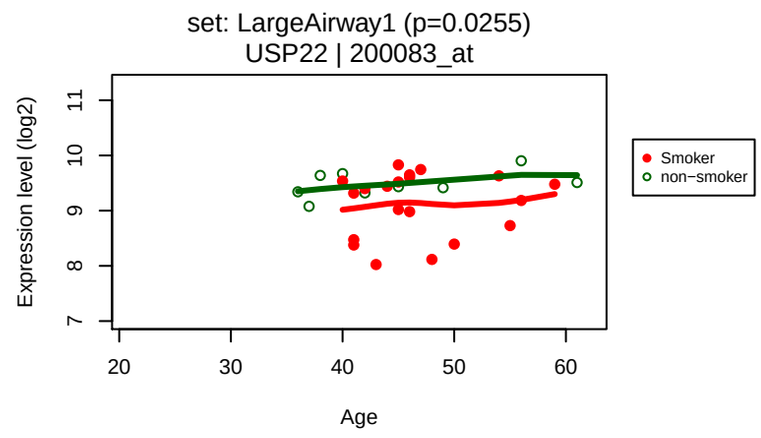

Gene expression of smokers and nonsmokers
gene: USP22 | 200083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0881
p=0.082
p=0.1373
p=0.0255
p=0.1494
p=0.6581
p=0.4714
p=0.1534
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

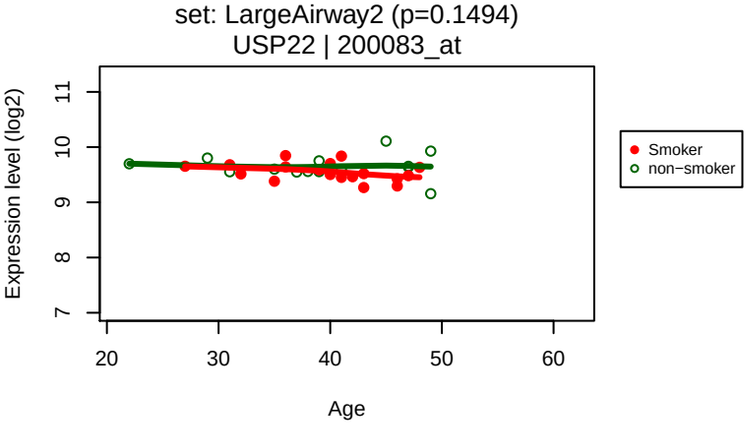

Gene expression ratio (smokers/nonsmokers)
gene: USP22 | 200083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0881
p=0.082
p=0.1373
p=0.0255
p=0.1494
p=0.6581
p=0.4714
p=0.1534
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

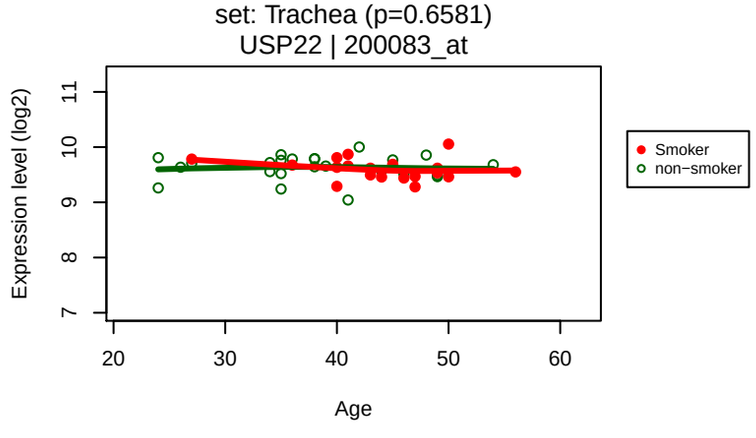

Gene expression ratio in Males
gene: USP22 | 200083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0322
p=0.9522
p=0.4188
p=0.1514
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP22 | 200083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9706
p=0.0196
p=0.3348
p=0.7329
SmallAirway2
LargeAirway2
Trachea
Macrophage2

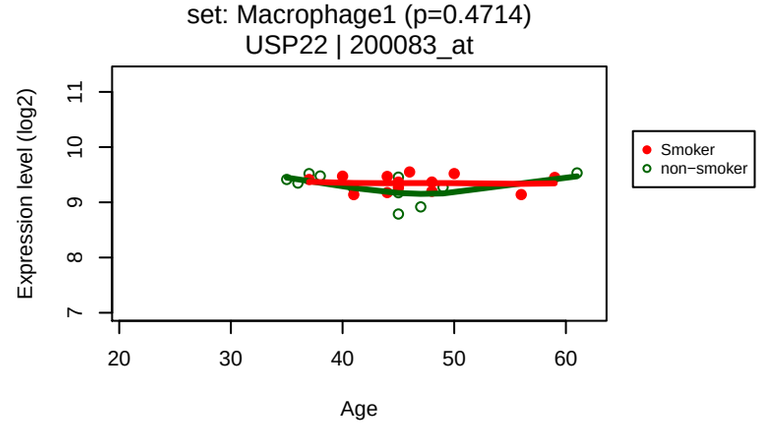

set: SmallAirway1 (p=0.0881)
USP22 | 200083_at
Expression level (log2)
Age
Smoker
non−smoker

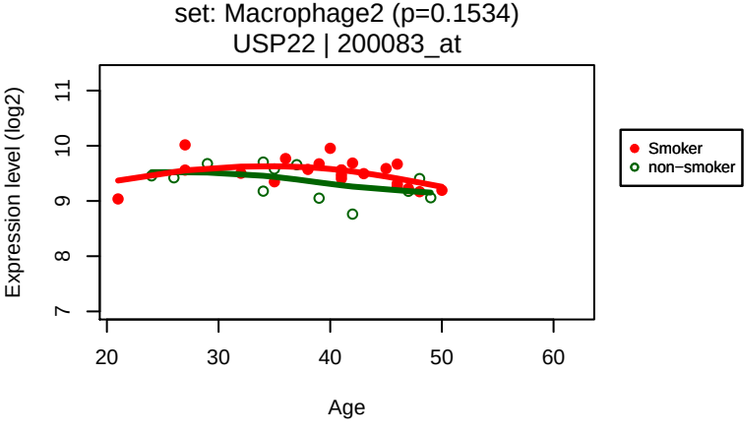

set: SmallAirway2 (p=0.082)
USP22 | 200083_at
Expression level (log2)
Age
Smoker
non−smoker

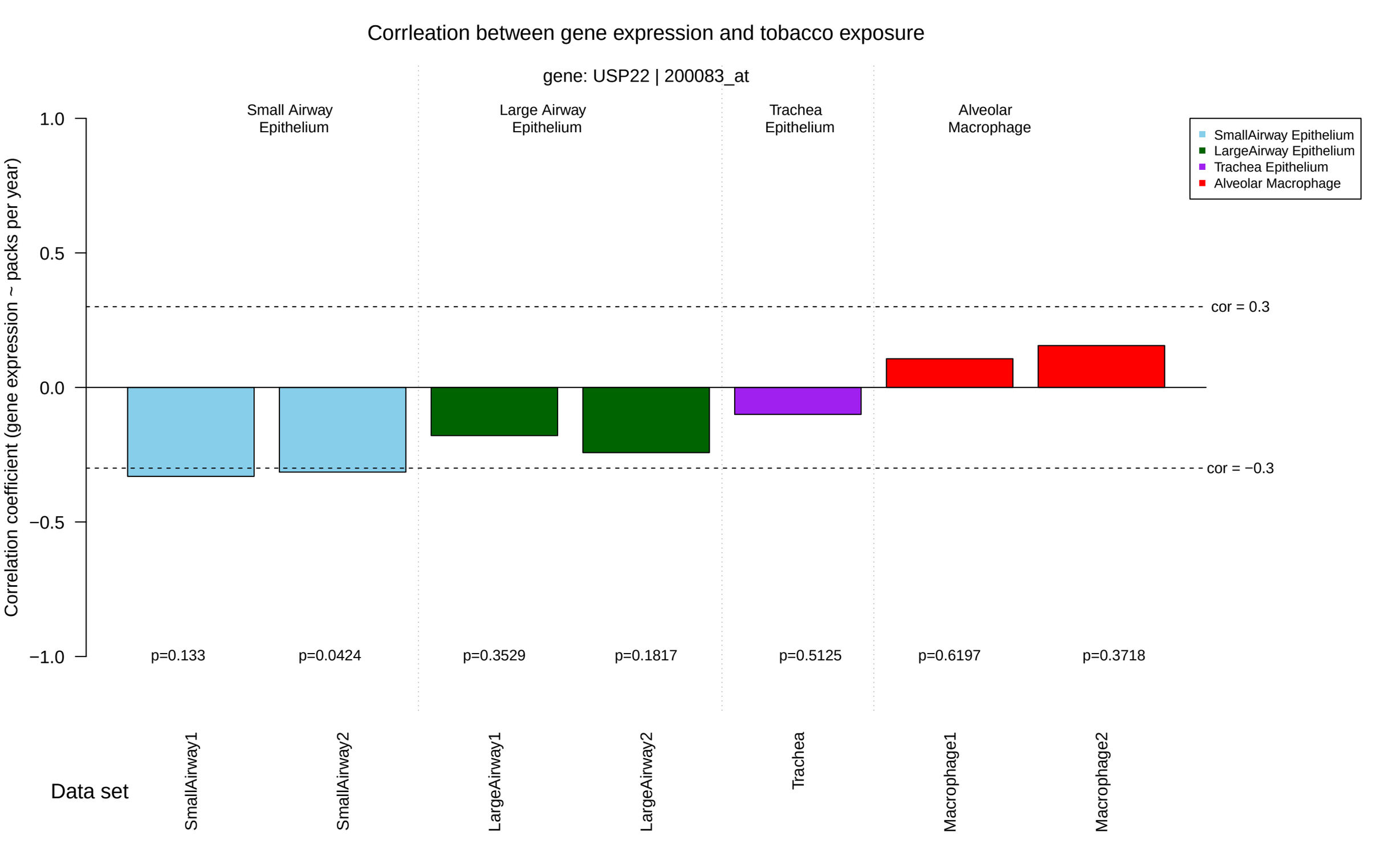

set: LargeAirway1 (p=0.0255)
USP22 | 200083_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1494)
USP22 | 200083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6581)
USP22 | 200083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4714)
USP22 | 200083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1534)
USP22 | 200083_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP22 | 200083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.133
p=0.0424
p=0.3529
p=0.1817
p=0.5125
p=0.6197
p=0.3718
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2